

Electronics and

Prices checked June 12, 2026 |

Metric	Value
Partial priced subtotal	\$653.39
BOM line items	23
Rows without complete USD cost	11

Pricing Notes

The subtotal includes only items with a numeric required quantity and a USD unit price.

Representative prices are assumptions and must be confirmed before ordering.

The CAROBOT shield is listed at CAD \$19.99 and is excluded from the USD subtotal.

Resolve the spreadsheet's 150:1 gearmotor description against linked Pololu item 34.

Sensor Bill of Materials

USD unless stated | Tax and shipping excluded

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93, which is 195:1.

Category	Component / Description	Supplier
Processing	Processor/controller board (exact model TBD)	Supplier TBD
Sensor	Adafruit VL53L0X time-of-flight sensor breakout, PID	Adafruit
Sensor	Adafruit TCS34725 color sensor breakout, PID 1334	Adafruit
Sensor	Adafruit BNO055 absolute-orientation sensor breakout, PID 2472	Adafruit
Sensor	Motor encoders; assumed Pololu 20D magnetic encoder pair kit, item 3499	Pololu
Vision	OpenMV Cam H7 Plus, SKU OMV-CAM-H7-PLUS-V1	OpenMV
Actuator	Pololu extended-shaft gearmotor; linked item 3493 is 195:1	Pololu
Actuator	28BYJ-style 5 V geared stepper motor; assumed Adafruit PID 858	Adafruit
Display	LCD screen (exact model and size TBD)	Supplier TBD
Lighting	LED strips (type/length TBD)	Supplier TBD
Mechanical	M4 suspension screws (length/type TBD)	Supplier TBD
Mechanical	Steel Flanged Bearing, Trade Number 74	McMaster-Carr
Mechanical	Motor brackets (exact model TBD)	Supplier TBD
Electronics	Protoboard; assumed Adafruit half-size Perma-Proto, PID 1609	Adafruit
Wiring	Qwiic cables; assumed 100 mm JST-SH 4-pin, Adafruit PID 4210	Adafruit
Safety/electronics	Circuit breaker (rating/model TBD)	Supplier TBD
Wiring	Wire clips (type and quantity TBD)	Supplier TBD
Wiring	JST-PH cables; assumed Pololu 6-pin female-female 10 cm, item 5643	Pololu
Power	Pololu D24V50F5 5 V step-down voltage regulator, item 2851	Pololu
Control	CAROBOT Motor Shield V3, SKU 2337	Canada Robotix
Control	SparkFun Qwiic Mux Breakout - 8 Channel (TCA9548A), BOB-16784	SparkFun
Power	LiPo battery (voltage, capacity, and model TBD)	Supplier TBD
Assembly	Robot mechanical structure and wiring assembly	Team designed and assembled

Status	Type	Required Qty	Existing Qty
Reused; existing inventory	Kit/off-the-shelf	0	2
Reused; existing inventory	Kit/off-the-shelf	7	14
Reused; existing inventory	Kit/off-the-shelf	1	2
Reused; existing inventory	Kit/off-the-shelf	1	2
Reused; spare inventory; installed quantity unknown	Kit/off-the-shelf	2	
Reused; existing inventory	Kit/off-the-shelf	2	4
Reused; existing inventory	Kit/off-the-shelf	4	8
Reused; existing and spare inventory	Kit/off-the-shelf	1	2
Reused; existing and spare inventory	Kit/off-the-shelf	1	1
Reused; existing inventory	Kit/off-the-shelf	3	3
Reused; existing inventory	Kit/off-the-shelf	1	2
Reused; existing and spare inventory	off-the-shelf	2	4
Reused; existing and spare inventory	Kit/off-the-shelf	4	8
Reused; existing and spare inventory	Kit/off-the-shelf	1	2
Reused; existing inventory	Kit/off-the-shelf	10	18
Reused; existing and spare inventory	Kit/off-the-shelf	1	1
Inventory status TBD	Kit/off-the-shelf		
Reused; existing and spare inventory	Kit/off-the-shelf	4	8
Reused; existing inventory; verify possible school spares	Kit/off-the-shelf	1	1
Reused; quantity must be verified	Kit/off-the-shelf		
Reused; quantity must be verified	Kit/off-the-shelf		
Reused; quantity and condition must be verified	Kit/off-the-shelf		
Reused; verify current configuration	Custom-built	1	

Spare Qty	Unit Price (USD)	Required Cost	Pricing Basis
1			TBD
1	\$14.95		\$104.65 Exact
0	\$7.95		\$7.95 Listed; discontinued
0	\$34.95		\$34.95 Exact
1	\$9.95		\$19.90 Representative
0	\$135.00		\$270.00 Exact; sold out
1	\$32.95		\$131.80 Linked model
4	\$4.95		\$4.95 Representative; out of stock
1			TBD
0			TBD
0			TBD
5	\$11.16		\$22.32 TBD
6			TBD
1	\$4.50		\$4.50 Representative
0	\$0.95		\$9.50 Representative
3			TBD
			TBD
2	\$3.23		\$12.92 Representative
0	\$29.95		\$29.95 Exact
			CAD price only
	\$6.95		Exact
			TBD
			TBD

Assumptions / Follow-up

Exact model required

PID 3317

Discontinued listing

PID 2472

Quantity is pair kits; confirm installed model

Production-run order only

Spreadsheet says 150:1; confirm ratio/model

Confirm exact model

Exact model and size required

Type, voltage, density, and length required

Price is per required set; specification required

Size/model required

Exact model required

Confirm dimensions

Confirm lengths

Rating/model required

Type and quantity required

Confirm pin count and length

Item 2851

CAD \$19.99; USD price and required quantity

Required quantity TBD

Voltage, capacity, model, and quantity required

Material costs TBD

Source URL
https://www.adafruit.com/product/3317
https://www.adafruit.com/product/1334
https://www.adafruit.com/product/2472
https://www.pololu.com/product/3499
https://openmv.io/products/openmv-cam-h7-plus
https://www.pololu.com/product/3493
https://www.adafruit.com/product/858
https://www.mcmaster.com/57155K437/
https://www.adafruit.com/product/1609
https://www.adafruit.com/product/4210
https://www.pololu.com/category/361/6-pin-jst-ph-style-cables
https://www.pololu.com/product/2851
https://www.canadarobotix.com/products/2337
https://www.sparkfun.com/sparkfun-qwiic-mux-breakout-8-channel-tca9548a.html